Ultrassonografia para TVP simplificada: Guia passo a passo

pocus101.com/dvt-ultrasound-made-easy-step-by-step-guide

Por favor ,

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A trombose venosa profunda (TVP) não diagnosticada ou não tratada nos membros inferiores é uma grande preocupação, pois um coágulo pode se desprender e causar embolia pulmonar e instabilidade hemodinâmica.

O ultrassom de compressão à beira do leito é uma maneira rápida e não invasiva de avaliar a TVP com altíssima sensibilidade e especificidade, permitindo determinar se a terapia anticoagulante é necessária (Burnside 2008).

Neste post, mostraremos como usar o ultrassom à beira do leito (POCUS) para:

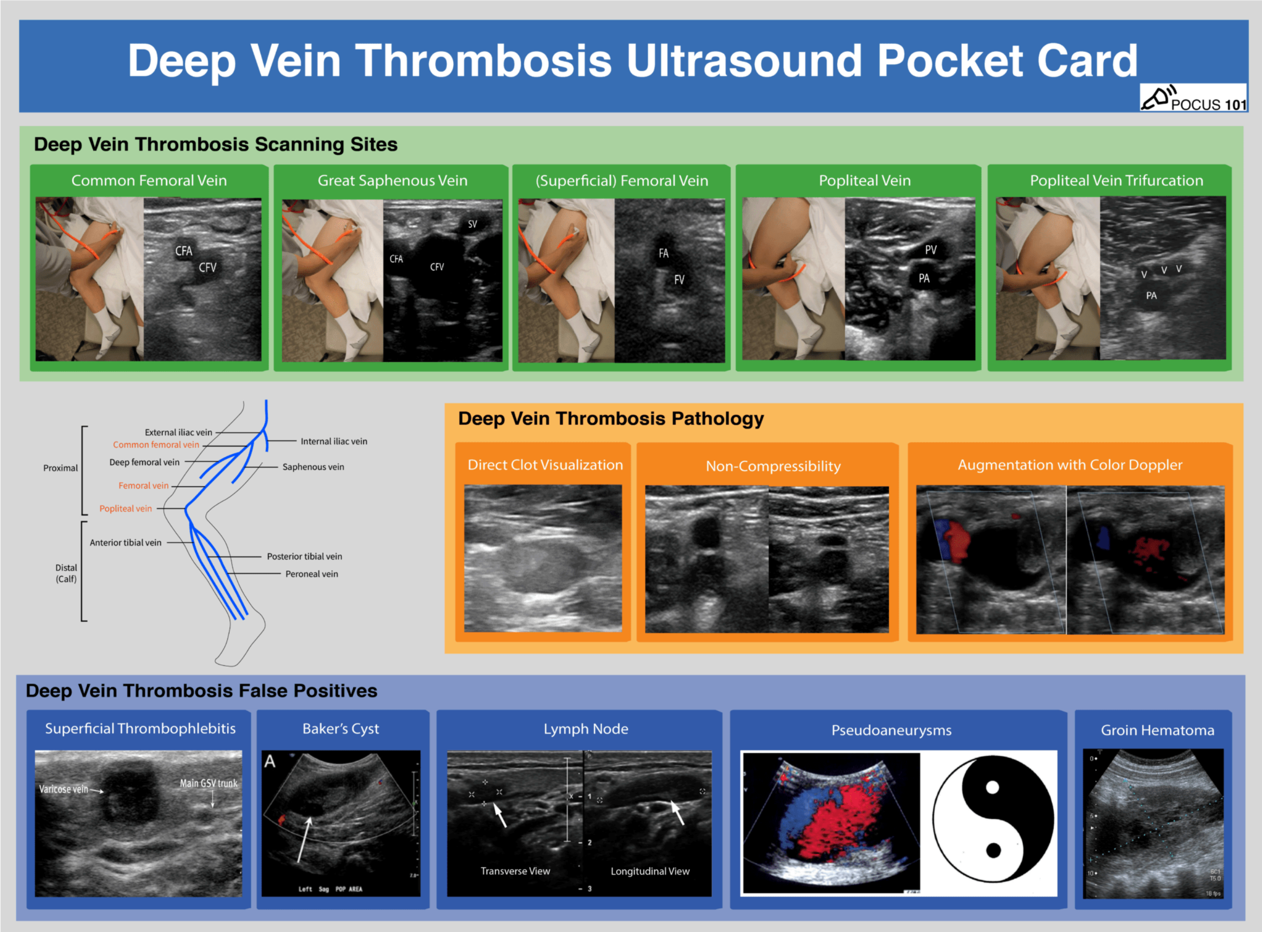
1. Realize o exame de ultrassom para trombose venosa profunda (TVP) dos membros inferiores passo a passo.
2. Avalie a presença de trombose venosa profunda (TVP) utilizando compressão, visualização do coágulo e Doppler colorido.
3. Identificar falsos positivos em ultrassonografias de trombose venosa profunda (TVP).

Após aprender esses princípios, você poderá usar o ultrassom de compressão à beira do leito para avaliar tromboses venosas profundas (TVP) nos membros inferiores com facilidade! Abaixo, você encontrará um **cartão de bolso em PDF sobre ultrassom para trombose venosa profunda** que pode baixar para ajudá-lo a lembrar os pontos mais importantes.

Índice

Cartão de bolso em PDF com informações sobre ultrassom para trombose venosa profunda

Baixe o PDF do cartão de ultrassom de TVP AQUI



[Cartão de bolso em PDF sobre ultrassom de trombose venosa profunda (TVP)](#)

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Indicações e contraindicações

Indicações de ultrassom para trombose venosa profunda

- Sintomas unilaterais nos membros inferiores, incluindo:
  - Dor na perna
  - Ternura
  - Inchaço
  - Eritema
- Sinal de Homans positivo (dor à compressão da panturrilha durante o exame físico)
- Qualquer preocupação com trombose venosa profunda (TVP).

Contraindicação do ultrassom para trombose venosa profunda

Existe um risco teórico de deslocamento de um coágulo e consequente embolia pulmonar por compressão. Essa é uma complicação rara, mas algo que você deve ter em mente (Lockhart, Sheldon & Robbin).

Preparação para ultrassom de trombose venosa profunda

Preparação do Paciente

- Peça ao paciente para deitar-se em decúbito dorsal.

  Opcional: Eleve a cabeceira da cama a 30°. Isso pode ajudar o sangue a se acumular nas veias dos membros inferiores e facilitar a visualização da veia.
- Gire externamente a perna do paciente e dobre o joelho na posição conhecida como " **perna de rã"** . Esta é a posição mais popular porque dilata a veia e a aproxima do campo de visão do plano da sonda de ultrassom. Além disso, a posição de perna de rã permite examinar a veia femoral comum até a veia poplítea distal sem precisar reposicionar o paciente (Read, Holdgate & Watkins).
- Coloque uma almofada sob o joelho do paciente para aumentar o conforto.

Posição de Pernas de Rã

Preparação do aparelho de ultrassom

- **Transdutor** : Sonda ultrassônica linear
- **Pré-configuração :** Venosa
- **Posicionamento do aparelho de ultrassom** : Posicione o aparelho no lado direito do paciente, de forma que você possa realizar o exame com a mão direita e operar os controles do ultrassom com a mão esquerda.

Posicionamento do paciente e da máquina

Anatomia da TVP por ultrassom

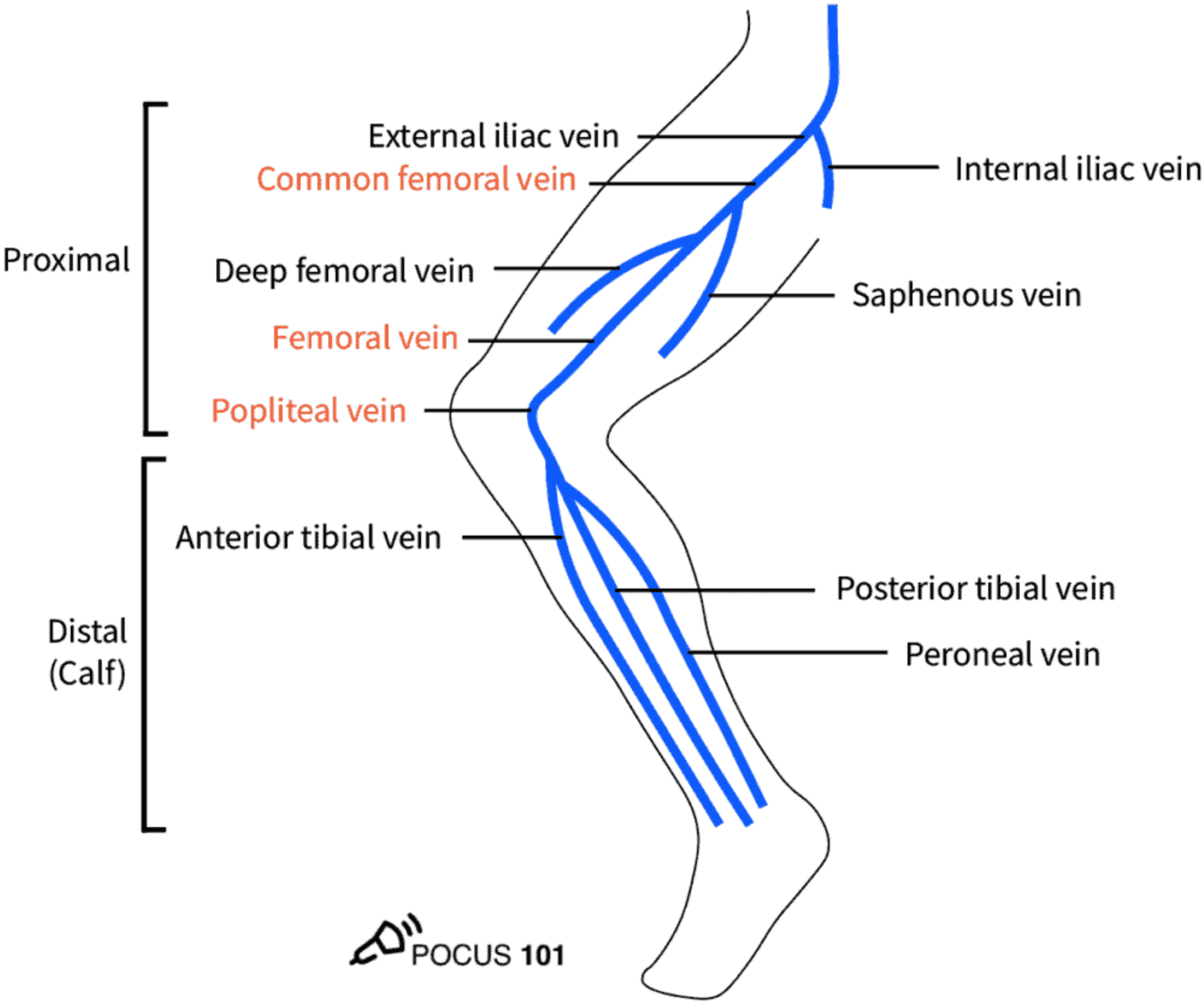
A anatomia venosa do membro inferior é relativamente simples. Indo de proximal para distal, aqui estão as veias mais importantes da perna que você precisa conhecer para o ultrassom de TVP (Trombose Venosa Profunda):

1. Veia femoral comum (VFC)
2. Veia Safena Magna
3. Bifurcação da VFC em Veia Femoral (anteriormente denominada Veia Femoral Superficial) e Veia Femoral Profunda
4. Veia poplítea
5. Trifurcação da veia poplítea
    1. Veia tibial anterior
    2. Veia tibial posterior
    3. Veia peroneal

The most proximal vessel is the **Common Femoral Vein (CFV)**, which gives off an important superficial branch known as the great saphenous vein. The CFV continues on to give off a deep femoral vein, and a **femoral vein**.

> *Editor's Note: The Femoral Vein is also previously known as the Superficial Femoral Vein. Don't be misled by the name – the (superficial) femoral vein is actually part of the* **deep venous system***. A study from 2011 showed that 75% of junior clinicians believed this vein was a part of the superficial venous system (Thiagarajah, Venkatanarasimha & Freeman). A clot in the (superficial) femoral vein is in fact a deep vein thrombosis and requires anticoagulant therapy to prevent a pulmonary embolus. Don't get confused by the nomenclature!*

As the femoral vein travels inferiorly and medially, it enters into the adductor canal and passes posterior to the knee to become the **popliteal vein**. At the level of the calf, the popliteal vein gives rise to three other deep veins: the **anterior tibial vein**, the **posterior tibial vein**, and the **peroneal vein** (Zitek, Baydoun & Baird).



Veins of the Lower Extremity

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Step-By-Step DVT Ultrasound Protocol

The 3-Point DVT Ultrasound Protocol

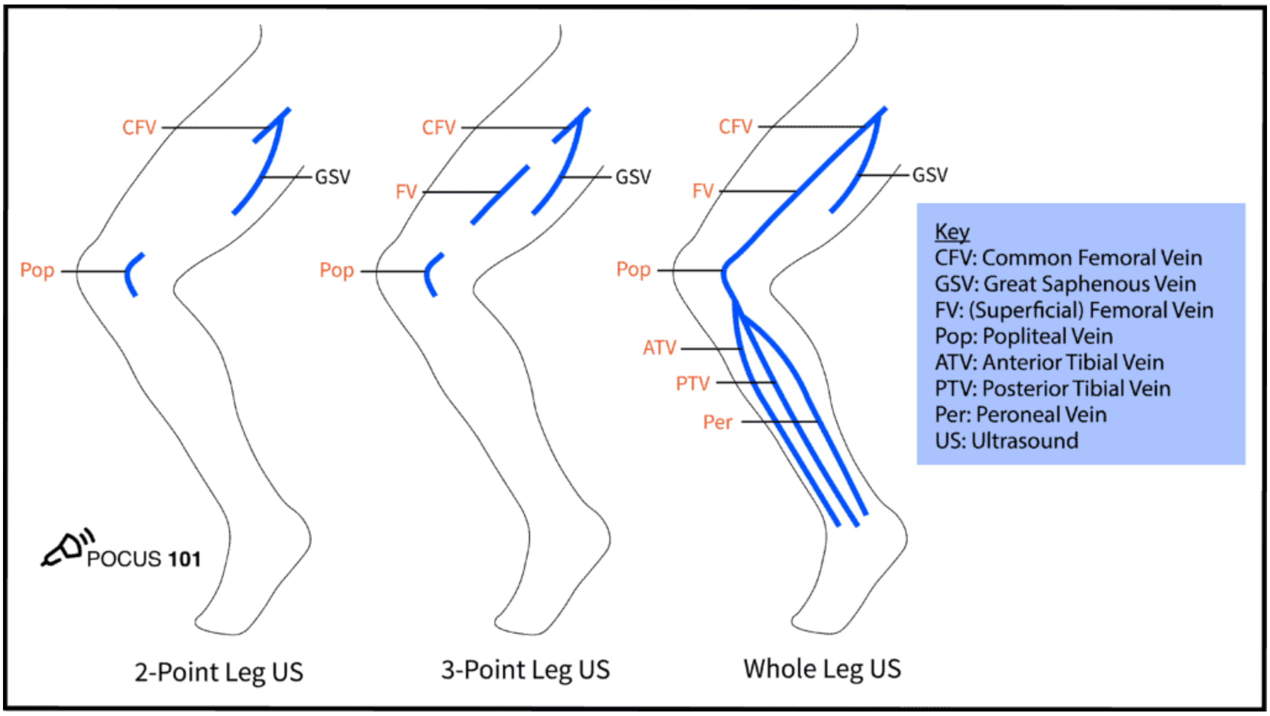
In this section, we will go over a step-by-step ultrasound approach to evaluate for deep vein thrombosis by visualizing the veins and using venous compression using a standard DVT Ultrasound Protocol.

It is important to note that there are several published protocols using anywhere from a 2-point ultrasound scan to a whole leg scan. We have found a 3-point compression ultrasound protocol to be most feasible while maintaining similar sensitivity and specificity when compared to a whole leg scan (Lee, Lee & Yun). We will focus on performing the 3-Point Ultrasound scan in this post, but here are the differences between the protocols that you may encounter:

2-Point Lower Extremity DVT Ultrasound: compression ultrasound including the femoral vein 1 to 2 cm above and below the saphenofemoral junction and the popliteal vein up to the calf veins confluence.

3-Point Lower Extremity DVT Ultrasound: compression ultrasound including the femoral vein 1 to 2 cm above and below the saphenofemoral junction, 1 to 2 cm above and below the bifurcation of the common femoral vein into the deep femoral vein and the (superficial) femoral vein, and lastly the popliteal vein up to the trifurcation into the anterior tibial vein, the posterior tibial vein, and the peroneal vein (Garcia 2018).

Whole Leg/Complete Lower Extremity DVT Ultrasound: involves scanning the leg using compression, color Doppler, and pulse wave Doppler from the common femoral vein all the way to the ankle while evaluating the calf veins.



DVT Protocols

Proper Vein Compression Technique

One of the biggest issues with performing DVT ultrasound is knowing how much to compress the vein. If the operator does not apply enough pressure or if the ultrasound probe is not perpendicular, these can lead to false positives for DVTs.

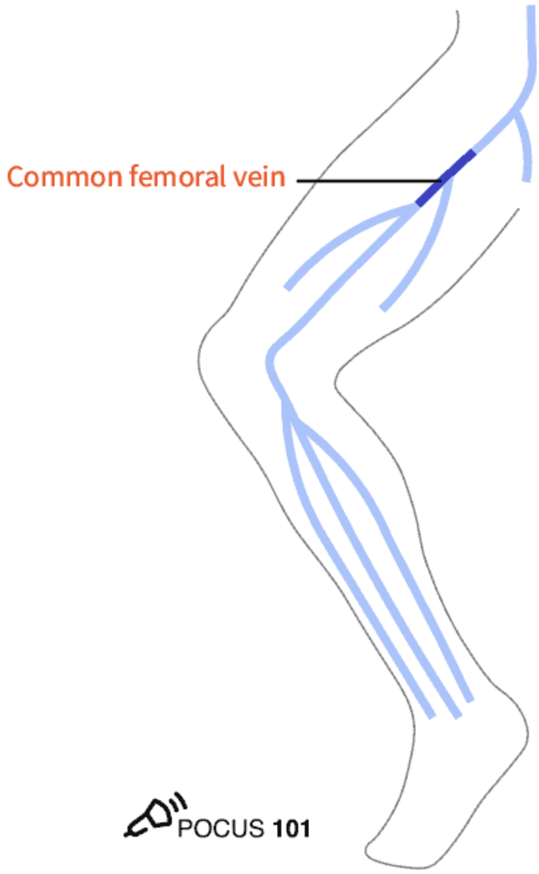
> You should apply pressure until the pulsatile artery compresses slightly. **If the adjacent vein compresses completely, there is no DVT.** Refer to the figure below for an example of proper technique. If the vein does not compress completely, it is likely the patient has a clot in that region.

Proper Vein Compression Technique

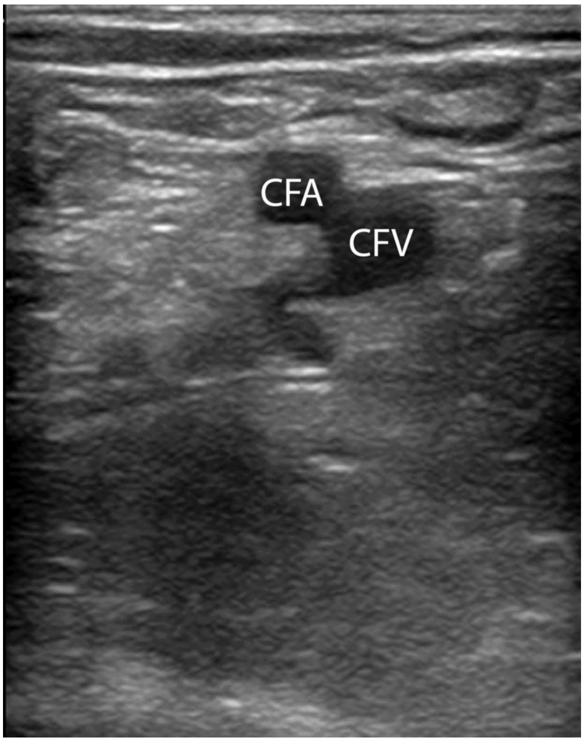
Editor's Note: There is a theoretical risk of dislodging a clot and causing a pulmonary embolism with compression. This is a rare complication, but something you should be aware of.

Step 1: Scan the Common Femoral Vein

- Apply gel onto the probe and place it along the inguinal ligament midway between the pubic symphysis and the anterior superior iliac spine (ASIS).
- Orient the probe **perpendicular** to the skin with the indicator facing the patient's right to obtain the transverse view.


Common femoral vein
POCUS 101

- Locate the common femoral vein and artery.
    - Notice the CFV is medial to the CFA.
- Apply firm pressure with the ultrasound probe until the artery compresses slightly. In a normal scan, the vein should compress entirely.

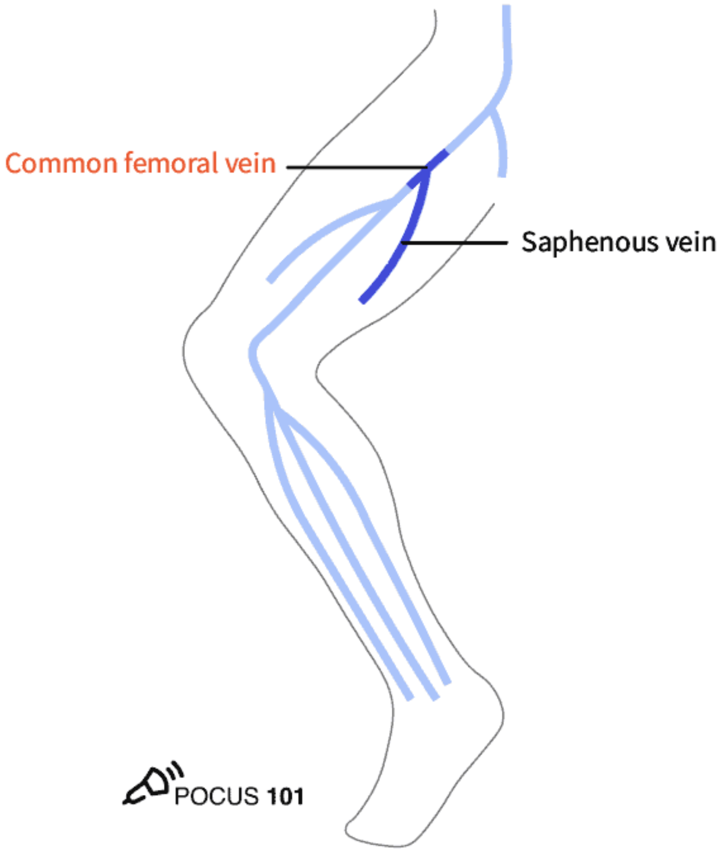


CFA: Common Femoral Artery; CFV: Common Femoral Vein

Compressible Common Femoral Vein

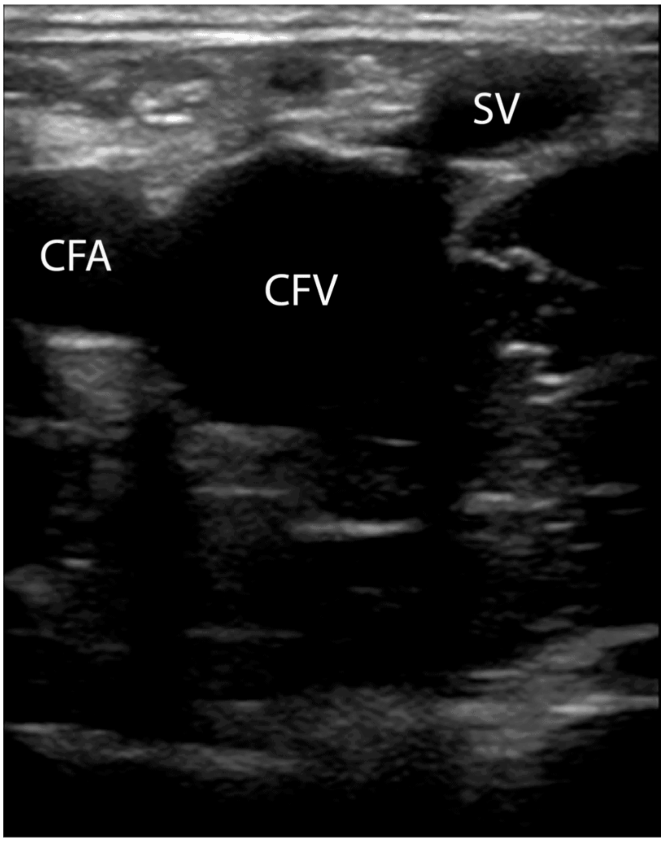
Step 2: Scan the Great Saphenous Vein

Slide the probe 1-2 cm down the patient's leg to find where the great saphenous vein branches off of the CFV.



- Keep the vessels centered in the image. As the probe moves distally, the artery will typically bifurcate first and then the vein.
- Compress the CFV at the junction with the Great Saphenous Vein.

- *Editor's Note: Depending on the size and proximity of a clot in the great saphenous vein to the CFV, there is evidence that these should be treated as DVT too.*

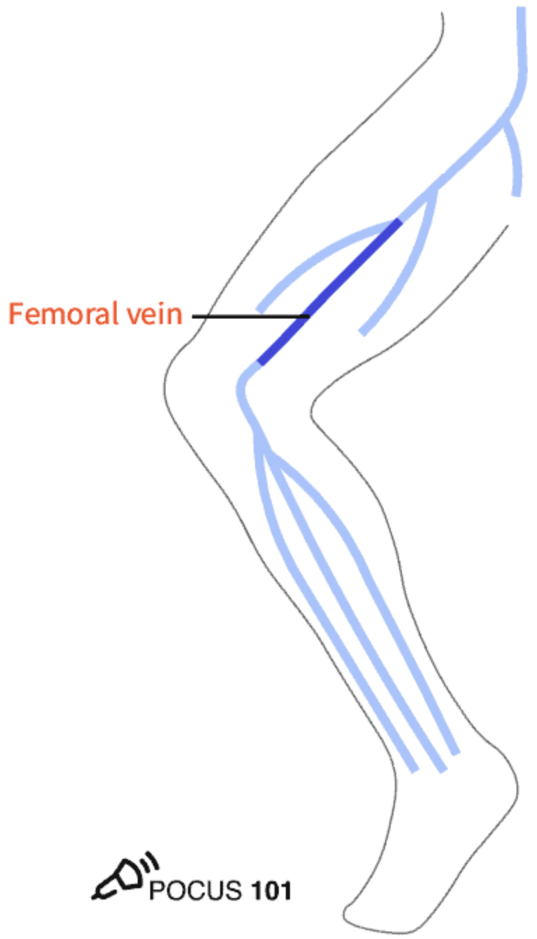


CFA: Common Femoral Artery; CFV: Common Femoral Vein; SV: Great Saphenous Vein

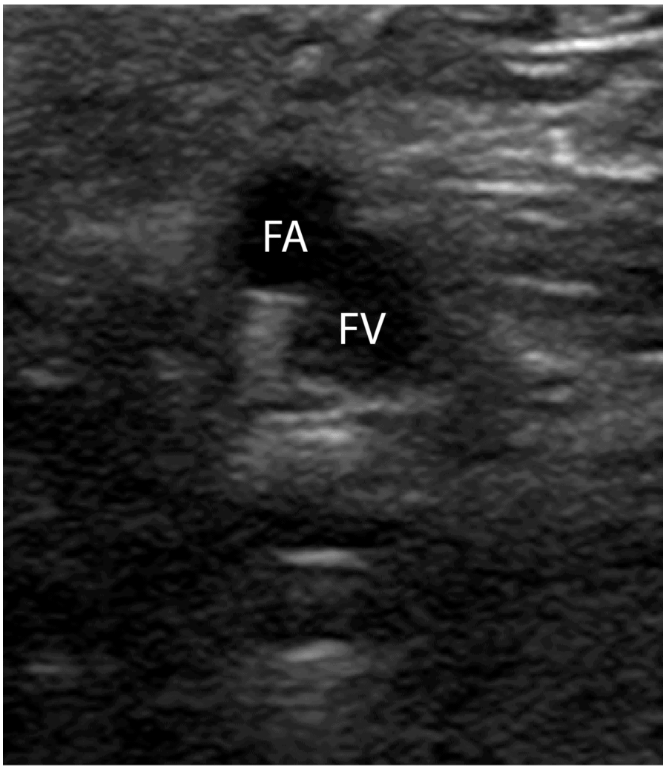
Compressible Great Saphenous Vein

Step 3: Scan the (Superficial) Femoral Vein

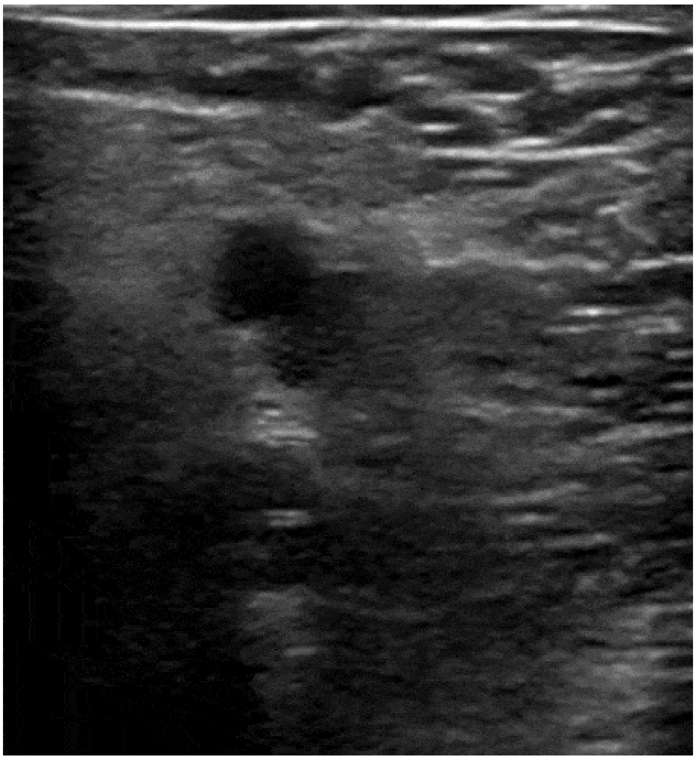
Slide the probe 1-2 cm down the patient’s leg to find where the CFV branches into the deep femoral vein and (superficial) femoral vein.



- At this point, the deep femoral vein will dive deep in the thigh. However, the (superficial) femoral vein will travel alongside the femoral artery.
- Compress the (superficial) femoral vein just distal to the bifurcation.
- *Optional: though the 3-point ultrasound protocol only requires you to compress just distal to the bifurcation, you can also periodically check for clots in the rest of the (superficial) femoral vein using compression as you gradually move the probe inferiorly and medially towards the popliteal fossa where the (superficial) femoral vein dives into the adductor canal.*



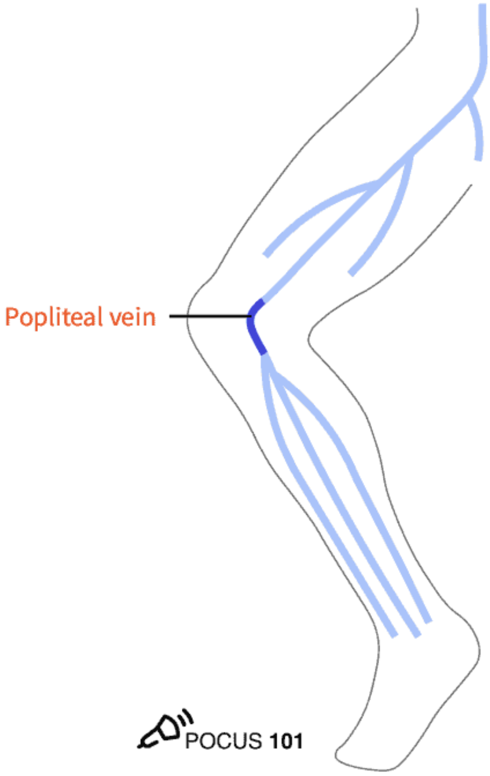
FA: Femoral Artery; FV: (Superficial) Femoral Vein



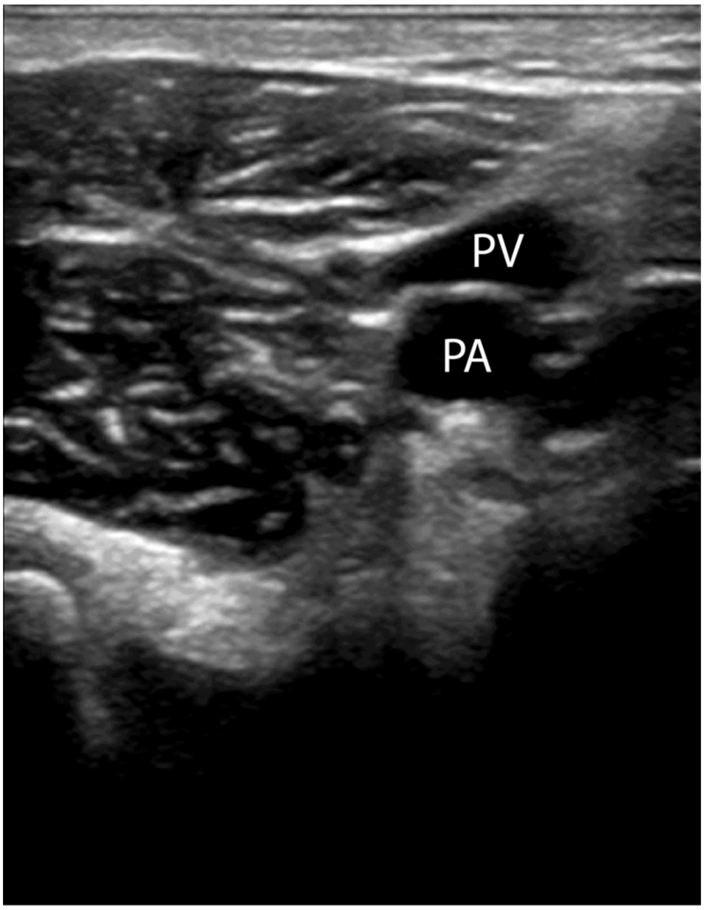
Compressible (Superficial) Femoral Vein

Step 4: Scan the Popliteal Vein

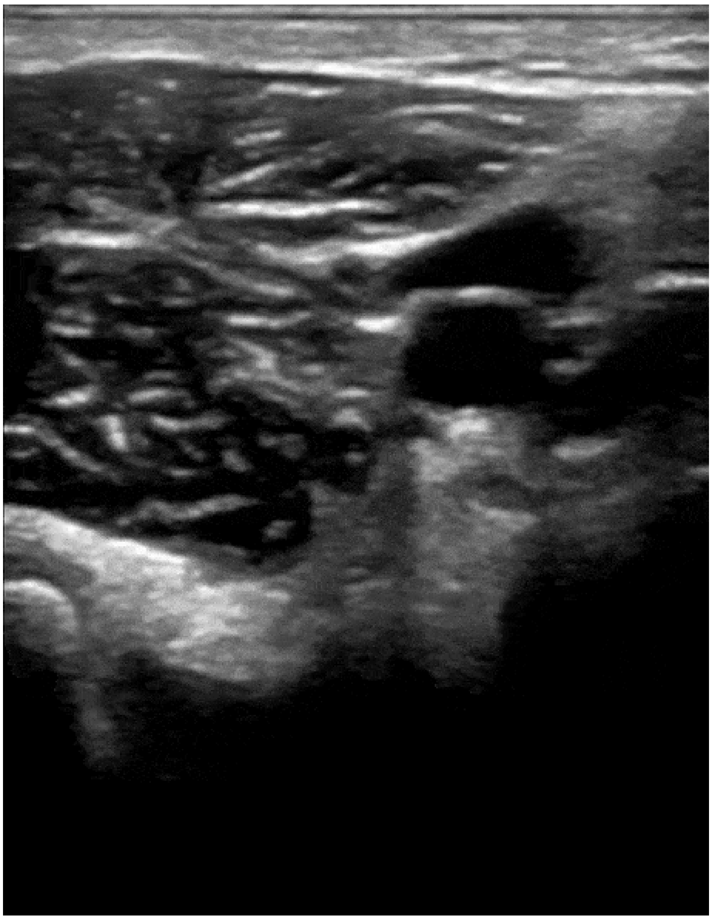
- Move the probe into the posterior crease of the knee and scan 2 cm above and below to find the popliteal vein.
- *POCUS 101 Tip: Sometimes it is difficult to locate the popliteal vein when you first start scanning. An easy way to locate it is to place the probe directly in between the two hamstring tendons behind the knee.*



- Use the probe to compress the popliteal vein and check for clots.
- *Editor's Note: Notice in this view that the popliteal vein is now on top and the popliteal artery is on the bottom. Think "**Pop on Top**!"*



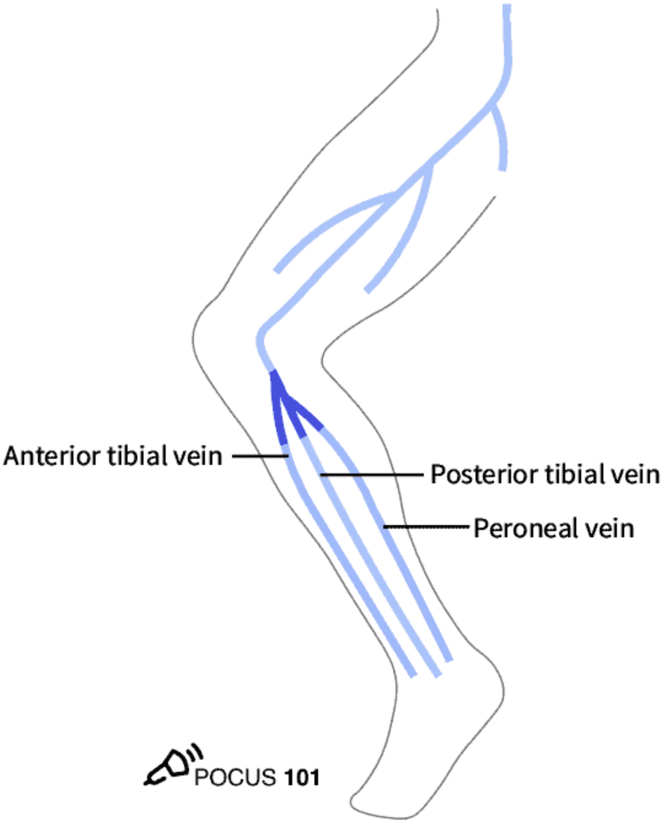
PA: Popliteal Artery; PV: Popliteal Vein



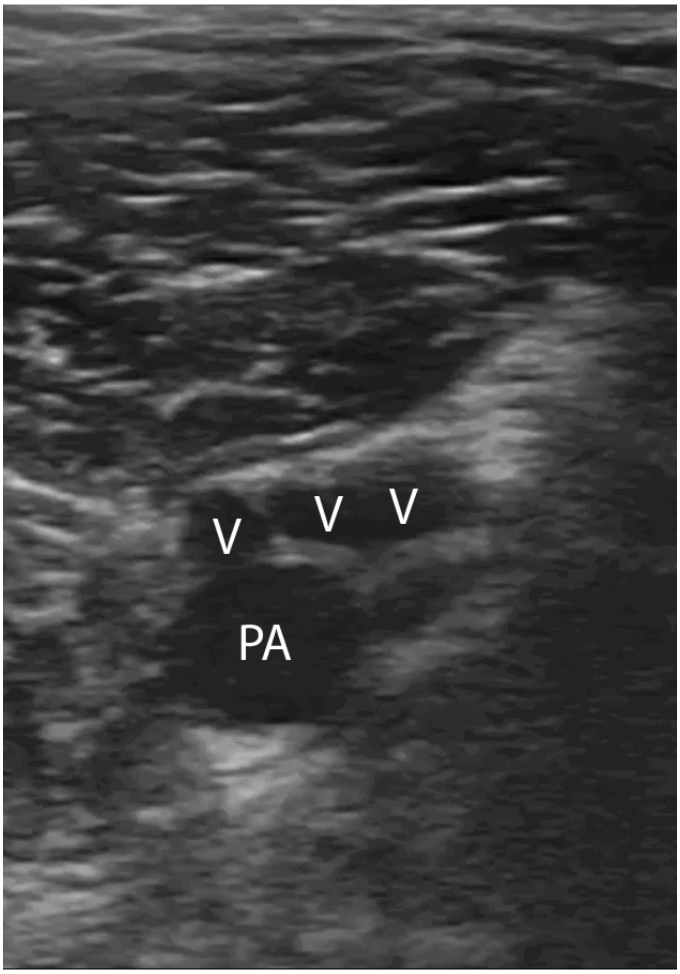
Compressible Popliteal Vein. Remember Pop on Top!

Step 5: Scan the Trifurcation of the Popliteal Vein

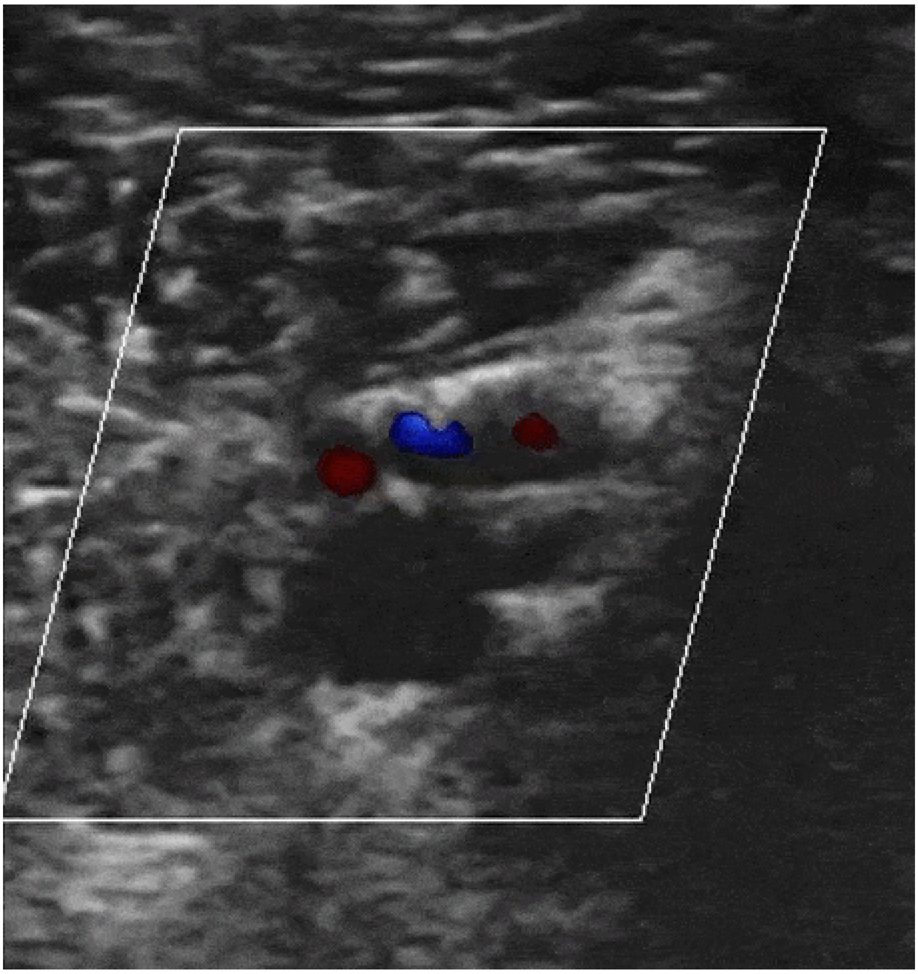
Continue to scan slightly more distal from the popliteal vein to find its trifurcation.



Compress the popliteal vein periodically until you find where the popliteal vein trifurcates into the anterior tibial, peroneal, and posterior tibial veins. This junction signals the end of the examination.



Popliteal Vein Trifurcation. PA: Popliteal Artery; V: Vein Trifurcation



Popliteal Vein Trifurcation Color Doppler

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Deep Vein Thrombosis Ultrasound Pathology Overview

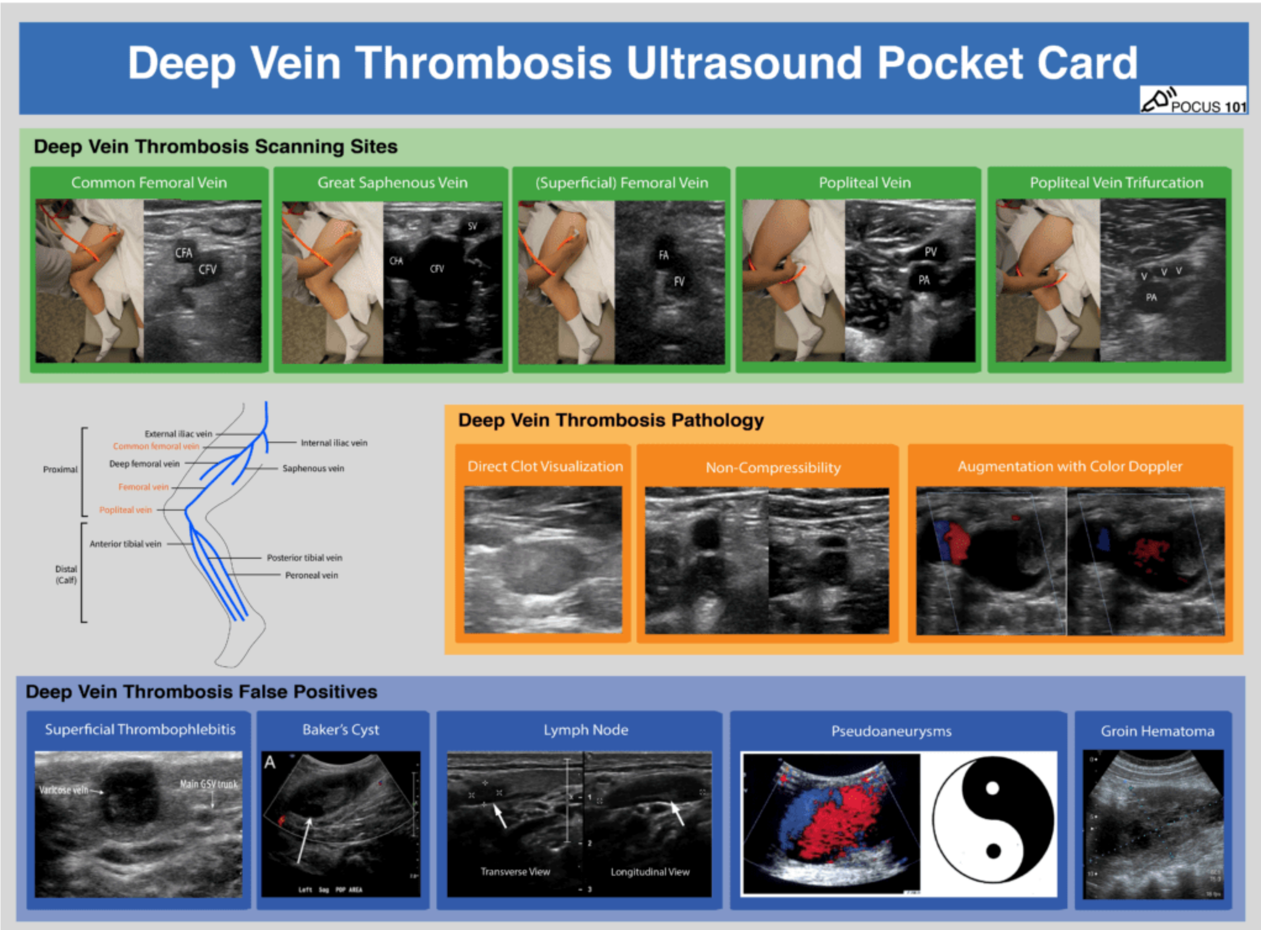
Deep vein thrombosis can be detected in three ways using point of care ultrasound: direct clot visualization, non-compressibility of vein, and augmentation with Color Doppler.

Direct visualization of the echogenic clot with ultrasound is a definitive way to detect a DVT. Among the indirect ways for diagnosing a DVT, compression ultrasound provides the highest sensitivity and specificity. A study of 220 symptomatic patients scanned for DVT with compression ultrasound yielded a 100% sensitivity and 99% specificity (Lensing et al, 1989). In comparison, augmentation with Color Doppler does not improve the accuracy of scanning and requires squeezing the patient's leg, which could dislodge any present clots and cause a pulmonary embolism (Lockhart, Sheldon & Robbin; Blaivas et al).

Our recommendation to diagnose DVT is to first use the direct visualization technique. If an echogenic mass is not apparent, we recommend evaluating the compressibility of the veins. The use of color Doppler and augmentation are optional modalities.

Remember to download the DVT Ultrasound Pocket Card PDF as a reference for DVT Ultrasound technique, pathology, and false positives:

Download DVT Ultrasound Pocket Card PDF HERE

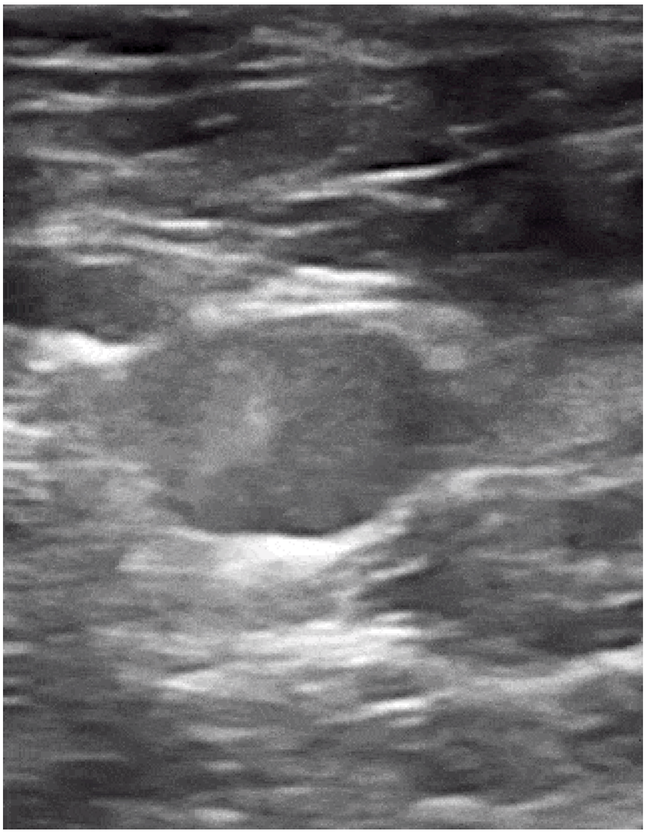


DVT Ultrasound Pocket Card PDF

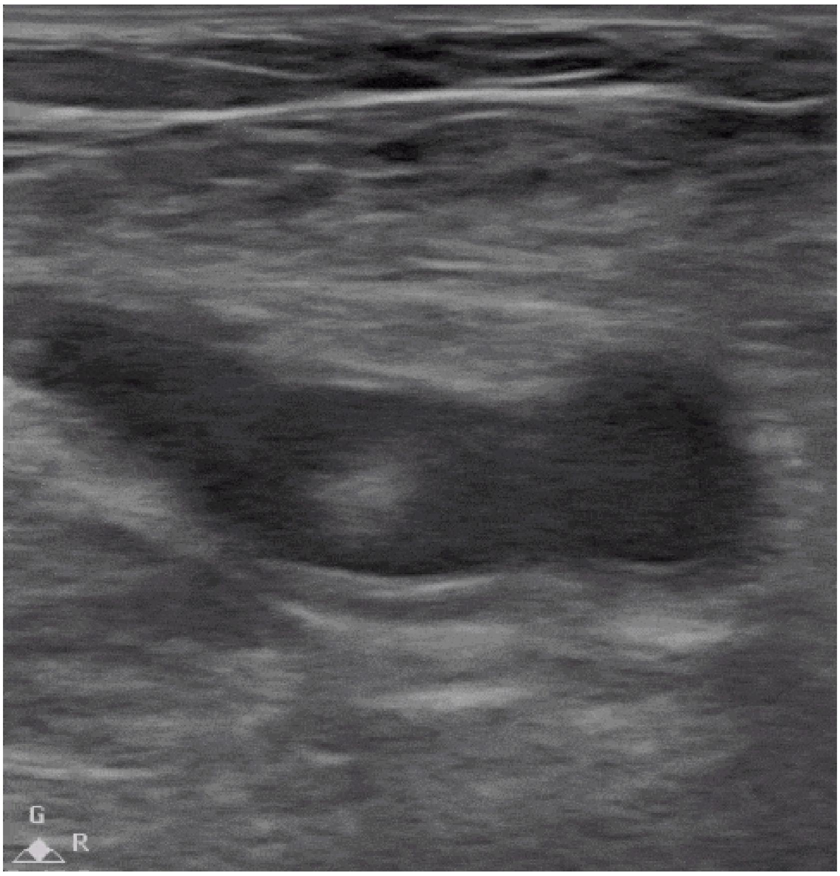
Direct Clot Visualization with Ultrasound

Depending on the age of a clot, you may see an echogenic mass inside the vessel on ultrasound if your probe is directly over the mass. You may not be able to see acute thrombi because they are anechoic or hypoechoic and they tend to be near the edges of the lumen. As the thrombus ages, it becomes more echogenic and retracts from the vein walls, making it much easier to detect with ultrasound. Therefore, when scanning for clots, do not just rely on direct visualization and always be sure to perform the compression technique to not miss any acute DVTs.

Examples of Direct Clot Visualization of DVT



DIrect Clot Visualization – DVT



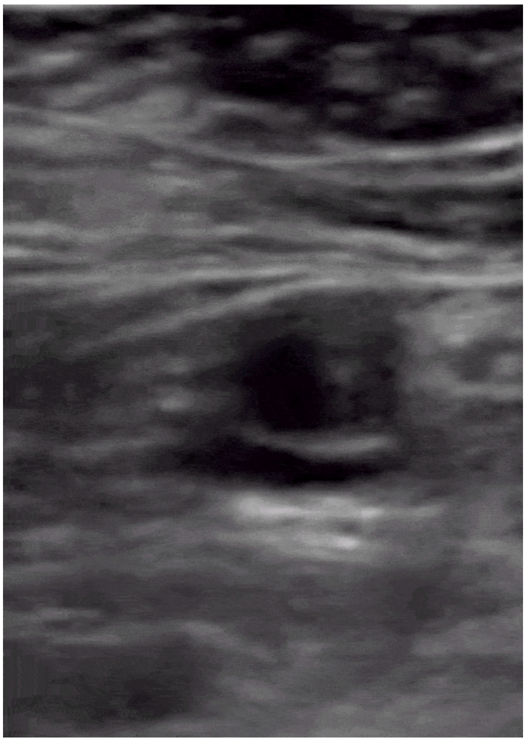
Echogenic Mobile Clot – DVT

Non-Compressibility with Ultrasound

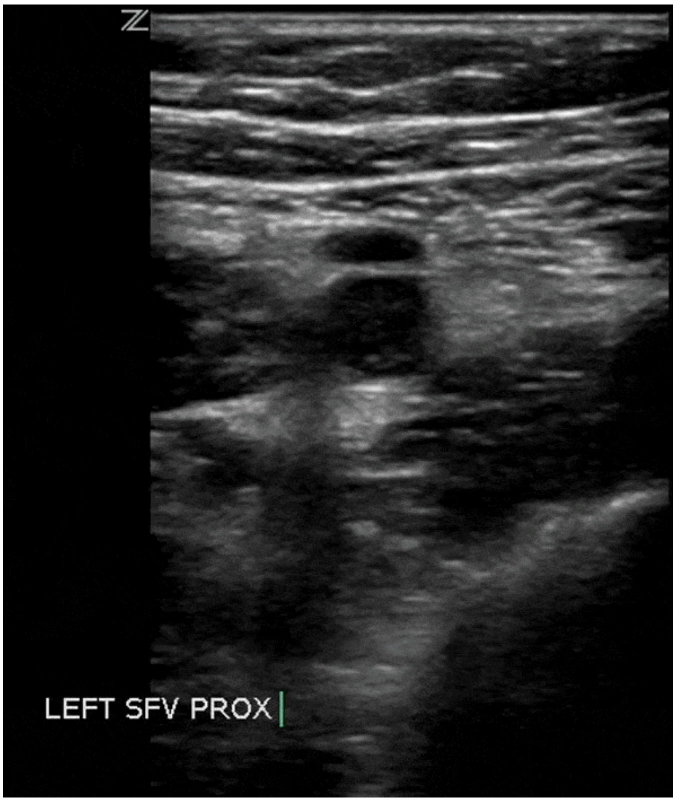
You can also detect a DVT by compressing the veins. If you apply pressure with the probe until the artery compresses slightly and the vein compresses completely, there is likely no DVT at that spot. However, if you apply enough pressure that the artery compresses but the vein does not compress completely this is considered a “non-compressible” vein and it is likely the patient has a clot in that region.

Examples of Non-Compressible Veins of DVT

Non-compressible Common Femoral Vein and Echogenic Clot



Non-compressible Popliteal Vein. Remember, Pop on Top!



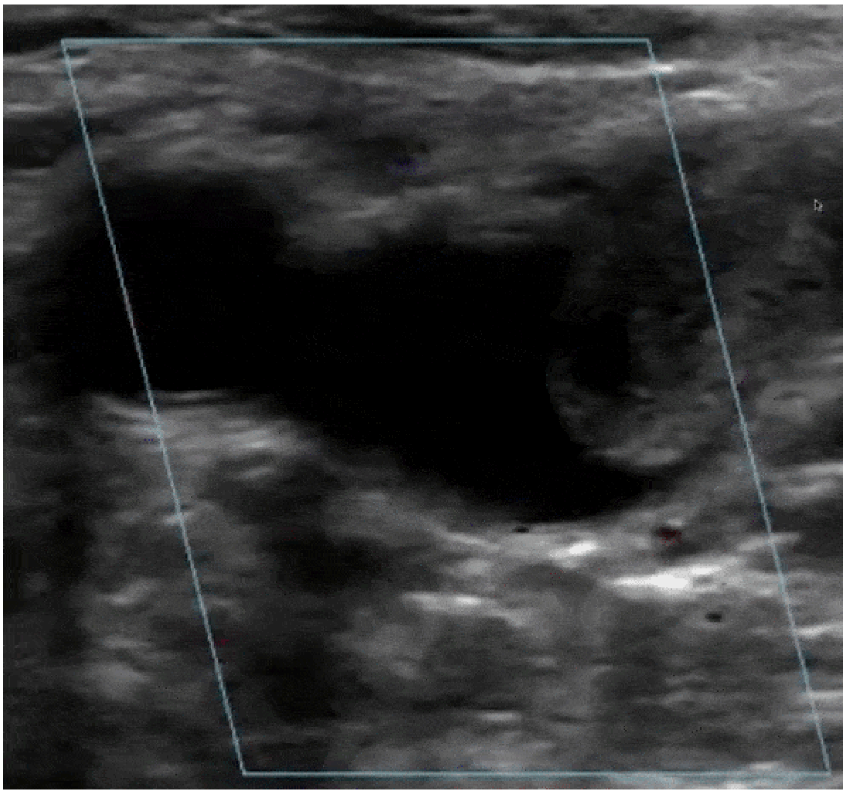
Non-compressible (Superficial) Femoral Vein

Augmentation with Color Doppler

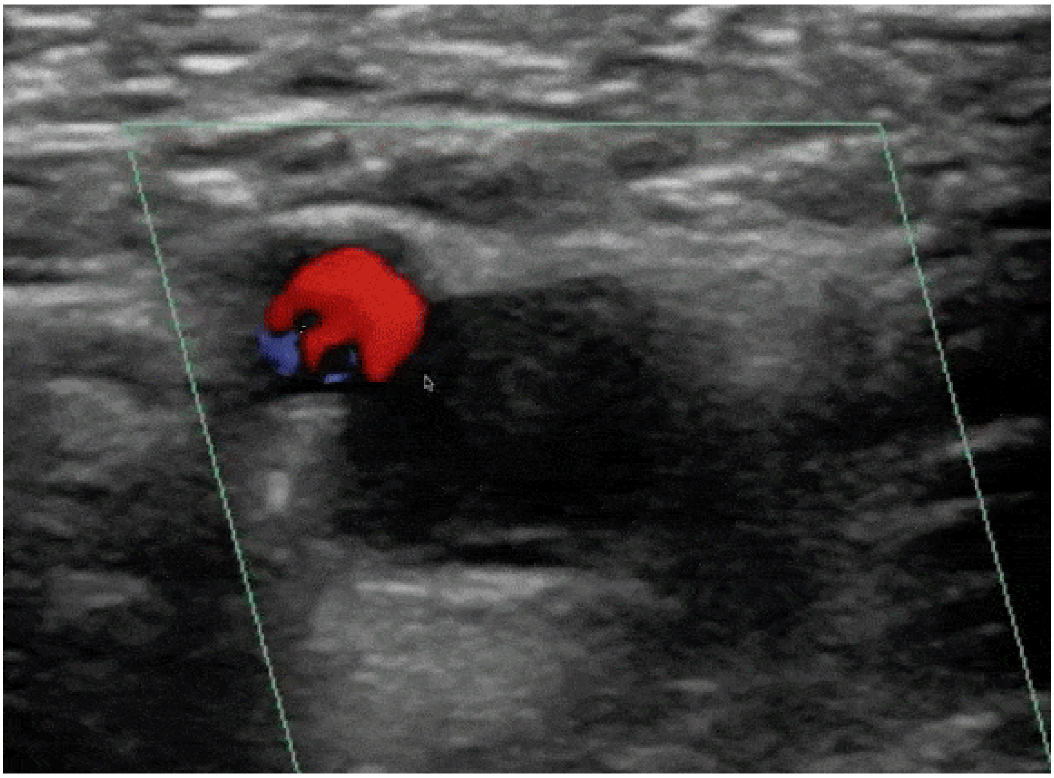
You can augment venous flow by squeezing the leg distal to where you are scanning. If Color Doppler demonstrates a homogenous increase in venous flow, this suggests that there is no occlusive thrombus between where you are squeezing and scanning. However, if there is no increase in venous flow, there could be a DVT.

Editor's Note: Use this method with caution as there could be a risk of dislodging any existing clots and causing a pulmonary embolism.

Obstructed Veins with Color Doppler of DVT



Clot in Great Saphenous Vein reduced Color Doppler flow



Obstructed (Superficial) Femoral Vein with no flow on Color Doppler

Video Summary for DVT Ultrasound

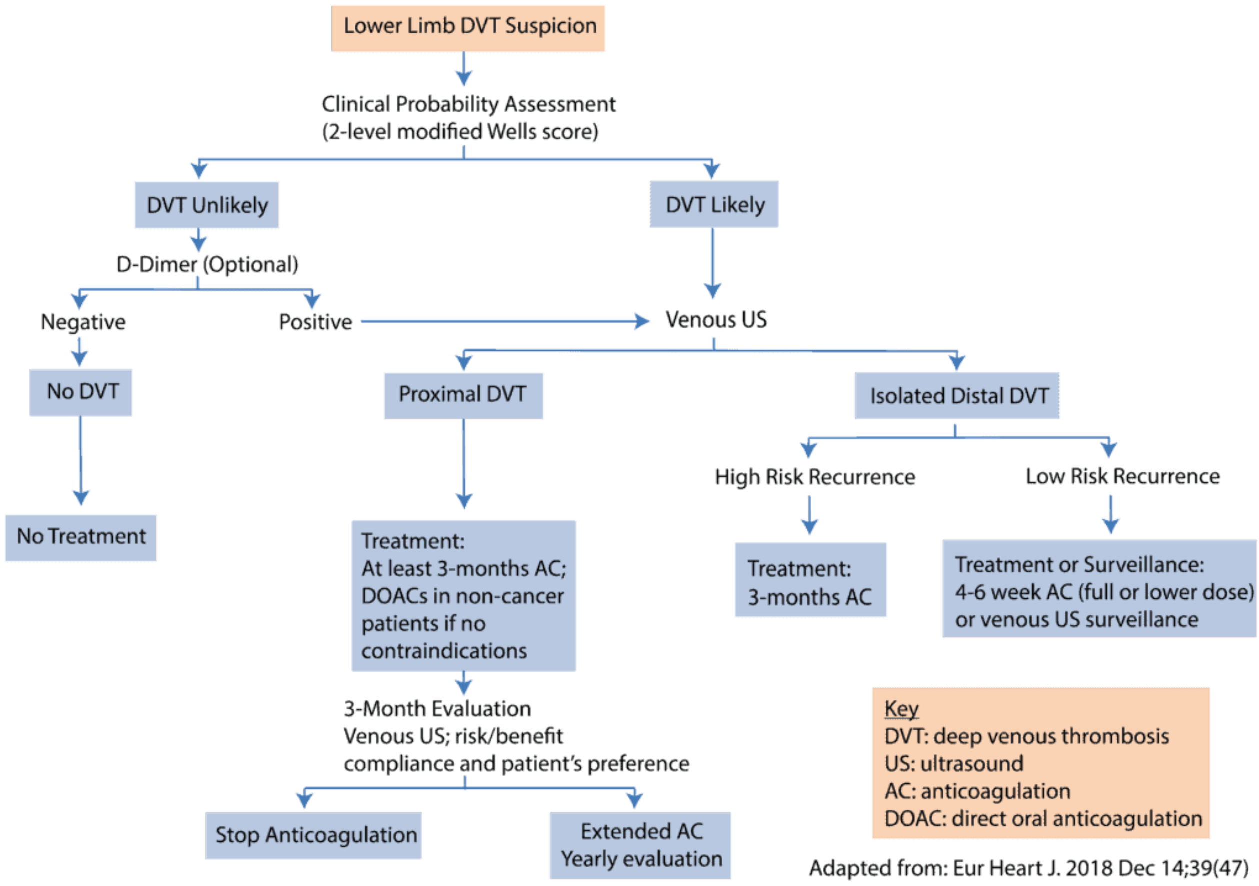
Below is a video we made summarizing how to obtain the views for DVT ultrasound as well as how to evaluate for DVT pathology:

[Watch on YouTube]

Deep Vein Thrombosis Treatment

It is important to treat DVT since one out of three patients with untreated DVT experience clinically significant pulmonary embolism, and the short-term mortality rate from untreated PE can exceed 20% (Burnside et al).

It is beyond the scope of this post to explain all of the possible treatments for DVT, but here is a proposed algorithm adapted from *Mazzolai* 2017:



DVT Management Algorithm

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Deep Vein Thrombosis False Positives

There are a few structures that can be mistaken for a deep vein thrombosis because they look like non-compressible vessels. These include superficial thrombophlebitis, Baker's cysts, lymph nodes, pseudoaneurysms, and groin hematomas, which we will show below.

To help avoid these false positives, be sure to always identify that the vein is next to an artery. Additionally, if you compare the vein in Short Axis and Long Axis views, it will look circular in the short axis view and cylindrical in the long axis view.

Circular Vein in Short Axis View

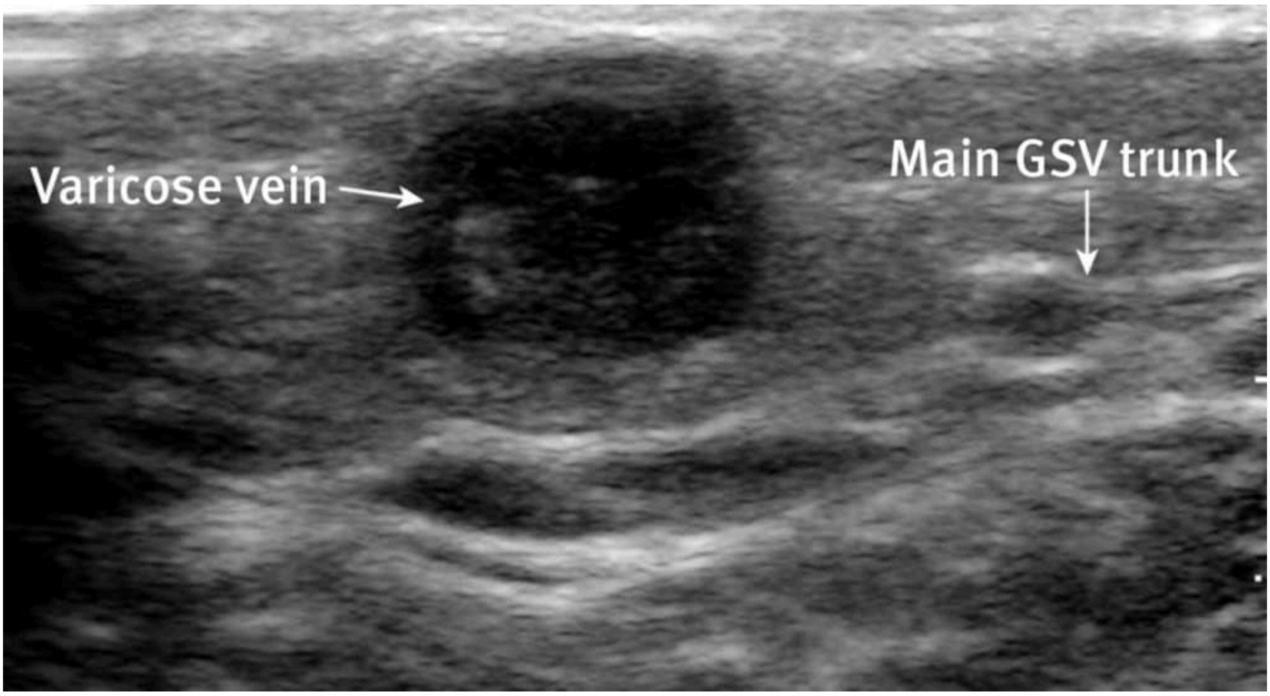
Cylindrical Vein in Long Axis View

Superficial Thrombophlebitis

Superficial thrombophlebitis occurs when there is a clot in a superficial vein, like the great and small saphenous veins and the varicose veins.

A clot in a superficial vein will show the same signs on ultrasound as in a deep vein; you can see an echogenic mass with direct visualization, non-compressibility, and decreased or absent color flow. The major difference is that superficial veins do NOT accompany arteries while deep veins do (Naringrekar et al).

This pitfall is more likely in obese patients whose superficial veins are enlarged and can be mistaken as deep veins.



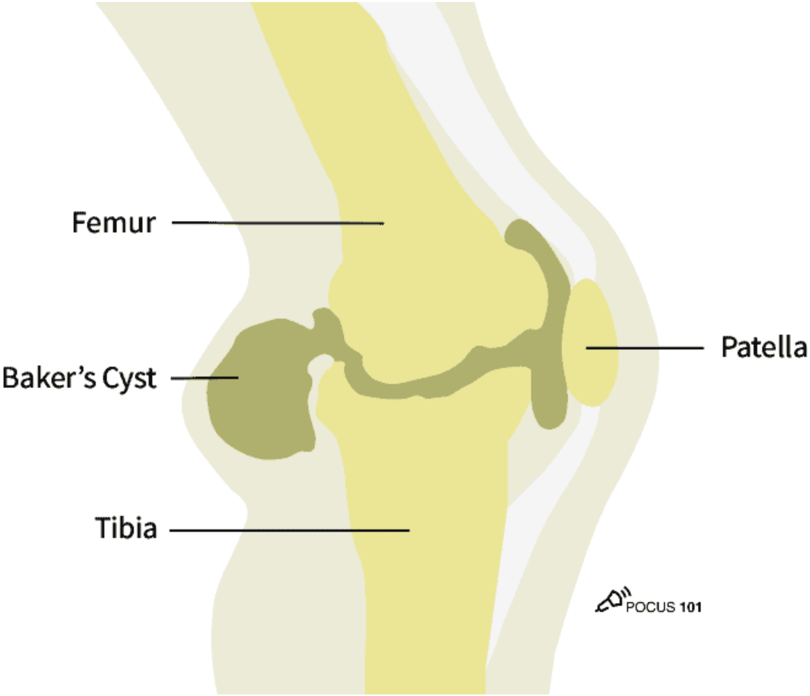
Superficial Thrombophlebitis (Nasr et al).

Baker’s Cyst

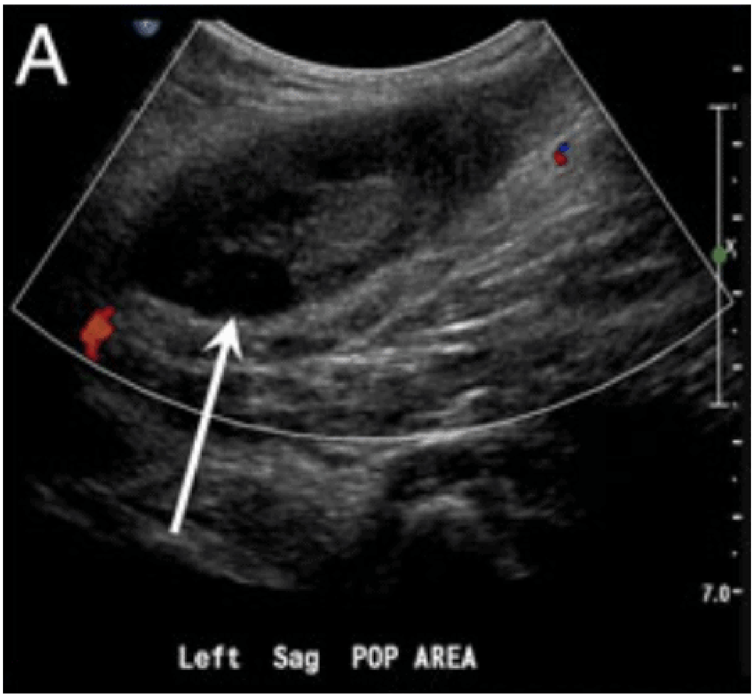
A Baker’s cyst is a fluid-filled cyst in the popliteal bursa on the back of the knee. This can result from inflammation of the knee joint or a knee injury, like a cartilage tear.

These popliteal cysts can present with similar symptoms as a deep vein thrombosis such as sharp knee pain, calf swelling, and occasionally erythema.

On ultrasound, a Baker’s cyst appears as a circular anechoic mass with sharply defined borders in boththe longitudinal and transverse view (Zitek, Baydoun & Baird). On Color Doppler, there should be no flow.



Baker's Cyst Illustration

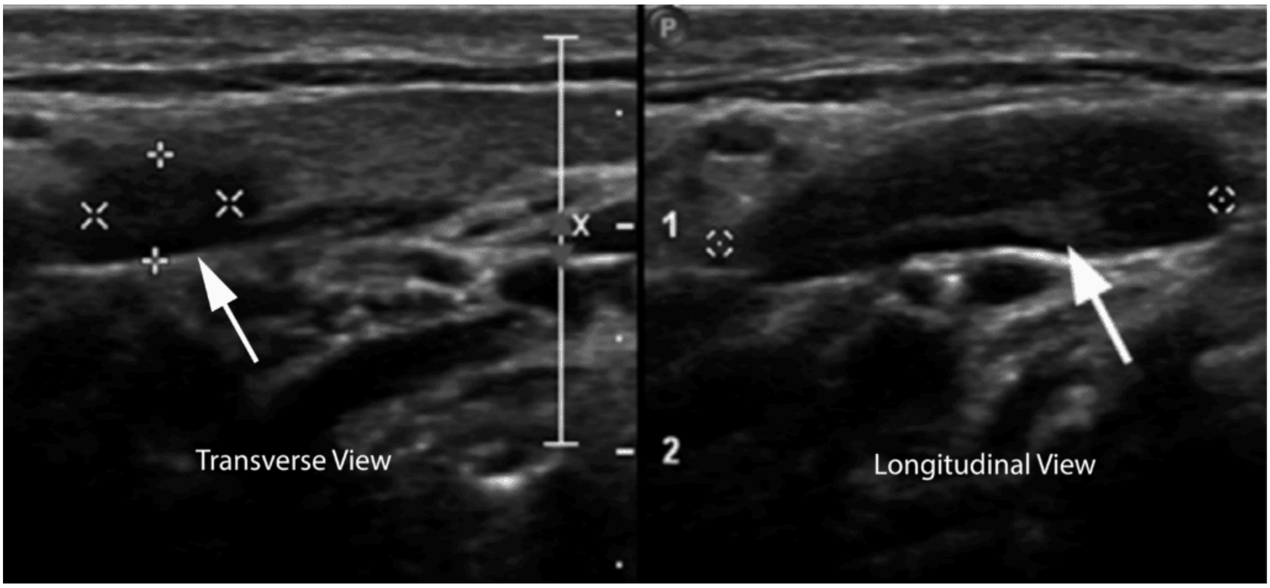


Baker's Cyst on Ultrasound (Naringrekar et al)

Lymph Nodes

Lymph nodes are part of the adaptive immune system and can become inflamed when the body is fighting an infection.

On ultrasound, a lymph node can look similar to a clot because it is a hypoechoic oval structure with a hyperechoic center. However, if you scan the lymph node in the longitudinal view, it becomes apparent that the object is circumscribed and not tubular in structure like a vein (Zitek, Baydoun & Baird). Lastly, a lymph node will disappear quickly if you attempt to trace it up and down, unlike an artery or vein that will be continuous structure.

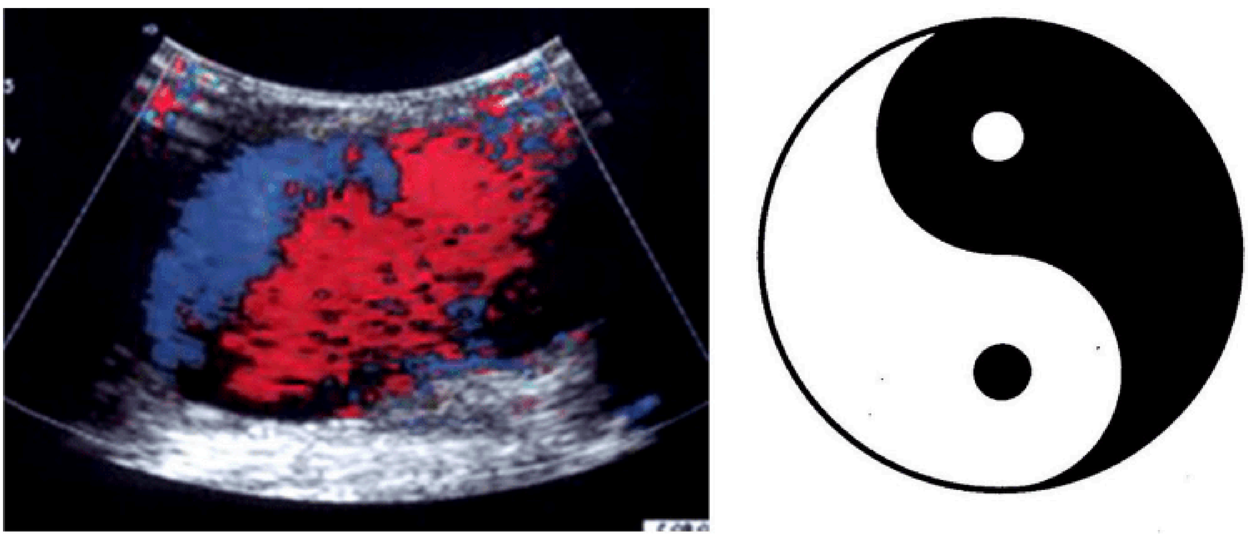


Lymph node (white arrows) in transverse and longitudinal view (Prativadi et al).

Pseudoaneurysms

A pseudoaneurysm occurs when there is a partial rupture of an artery wall that leads to an outpouching of the artery with the accumulation of periarterial blood. This most commonly occurs from arterial trauma from an interventional medical procedure, gunshot, or a stab wound (Montorfano et al). These sites can become infected and give off emboli.

On ultrasound, pseudoaneurysms present as anechoic or hypoechoic images, sometimes with moving echoes. With Color Doppler, you can see a **"Yin-Yang Sign"** from the circular motion of blood inside the pseudoaneurysm cavity. In systole, blood flows towards the pseudoaneurysm cavity (red) while in diastole the blood moves back towards the arterial lumen (blue)(Montorfano et al).

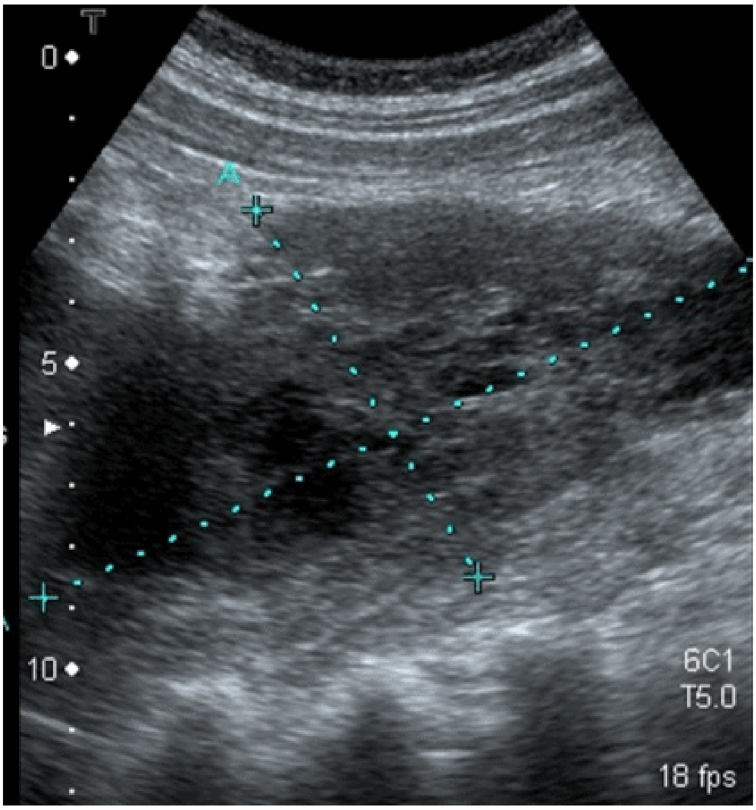


Pseudoaneurysm with Yin Yang Sign (Montorfano et al).

Groin Hematoma

A groin hematoma occurs when blood leaks from an artery, typically the femoral artery, and causes localized swelling. Some risk factors can include recent procedure in that area, obesity, anticoagulation, and peripheral vascular disease (Kosmidou & Karmpaliotis).

On ultrasound, a groin hematoma will be hypoechoic with some anechoic areas scattered throughout. If you scan the hematoma in the longitudinal view, it becomes obvious that the object is circumscribed and not tubular like a vein or artery.



Groin Hematoma (Ony et al).

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